CS-2007L Data Structures Lab — Course Delivery Design

Date: 2026-10-19
Course: Data Structures Lab (C#), CS-2007L, 1(0-3)
Institution context: DHA Suffa University (aligned to official 2026 outline + Lab Manual 1)

Goals

Build a complete teach + self-study package for the lab course:

- Student-facing English materials (manuals, lecture slides)
- Teacher prep (understanding slides, notes)
- Viva question slides
- Roman Urdu lecture/talking scripts (English technical terms kept)

Decisions (locked)

Decision	Choice
Role	Teaching + self-study
Package	Slides + lab manual + teacher notes + lecture scripts
Language	English (students) / Roman Urdu + EN terms (teacher scripts)
Syllabus fidelity	Same topics/order as official outline; polish pacing & tasks
Approach	Mirror + Upgrade (keep Lab 1 structure; upgrade delivery)
Build order	Lab 1 complete template first, then Labs 2–14

Per-lab deliverables (template)

```
CS-2007L/LabXX/
  slides/
    A_lecture.md                # projected in class
    B_teacher-understanding.md  # private prep
    C_viva-bank.md              # exit ticket / viva / quiz warm-up
  02_lab-manual.md
  03_teacher-notes.md
  04_lecture-script-roman-urdu.md
  starter/                      # optional skeleton
```

Three slide decks

1. **A — Lecture:** ~20–25 slides, one idea each, live demo cues, no code dumps.
2. **B — Teacher understanding:** why, pitfalls, hint ladders, marking traps (Roman Urdu ok).
3. **C — Viva bank:** question → model answer, CLO-tagged, easy→hard.

Lab 1 session (3 hours)

Time	Block
0:00–0:15	Hook + CLOs
0:15–0:35	Toolchain live
0:35–0:55	Debugger (Example 2)
0:55–1:15	Types + records (Ex 3–4)
1:15–1:35	Six ops (Ex 5)
1:35–1:50	Tradeoff (Ex 6)
1:50–2:40	Graded Tasks 1–4
2:40–2:55	Extension / homework pointer
2:55–3:00	Exit ticket + AI disclosure

Rollout after Lab 1

1. Freeze Lab 1 quality bar from this package.
2. Replicate same mould for Labs 2–14 per weekly outline.
3. Add mid/final/CCP packs later.

Out of scope (for now)

- PowerPoint/Google Slides binary files (markdown outlines first)
- Video recording
- Changing official CLO weights or week topic order